The Alternative Credit Bureau

How Vietnam's SMEs get credit they deserve

Shinhan Future's Lab Hackathon

The Problem

The Credit Gap

What Banks Want:

- 6 months statements
- Tax returns
- Collateral
- Formal business registration

What SMEs Have:

- Cash registers (POS)
- Shopee/Lazada orders
- TikTok Shop sales
- MoMo wallet transactions
- Daily sales receipts

The Reality

Creditworthy Businesses Getting Rejected

40.5%

of Vietnam's firms have bank loans

World Bank Enterprise Survey: Many SMEs underserved by traditional credit

The Data Reality:

- Street food vendor: **50+ transactions/day**
- TikTok seller: **200 orders/week**
- POS merchant: **Monthly sales in digital systems**
- **Existing solutions use ONE data source. We combine MULTIPLE.**

The Insight

Why do banks reject SMEs?

They look at the wrong data.

What data do SMEs actually generate?

- Sales receipts
- Order volumes
- Settlement patterns
- Customer returns

What if we used that data?

We'd see businesses banks can't see.

The Solution

Shinhan Credit Connect

We combine data sources banks ignore into a credit score that reflects actual business health:

- **POS transaction data** (SB9)
- **Marketplace sales** (SF12)
- **E-wallet cash flow** (SF12)
- **Settlement timing patterns**
- **Seasonal variations**
- **Return/refund behavior**

The Demo

From Transaction to Loan Offer

Step 1: Merchant installs SOL POS / connects marketplace

Step 2: Transaction data flows in over time

Step 3: Credit score starts forming

Step 4: After 90 days: pattern emerges

Step 6: Illustrative loan offer generated

Step 7: Funds: Decision in 72 hours

This is a mock scenario for demonstration. We show the CONCEPT of data → score → recommendation.

How It Works

Layer 1: POS Data (SB9)

- Daily sales volume
- Transaction count
- Average ticket size
- Refund rate
- Settlement timing (daily/weekly)
- Peak hours patterns

How It Works

Layer 2: Marketplace Data (SF12)

- GMV from Shopee/Lazada/TikTok
- Order completion rate
- Return rate
- Cancellation rate
- Response time
- Review scores

How It Works

Layer 3: E-Wallet Data (SF12)

- Cash flow patterns
- Inflow/outflow ratio
- Recurring payments
- Balance trends

How It Works

The Scoring Model

Signals We Analyze:

- Sales Consistency
- Growth Trend
- Cash Flow Health
- Return Rate
- Settlement History
- Seasonal Patterns

Weights:

- Weights are illustrative - to be validated with pilot data
- Model will be backtested against historical outcomes

This is a proof of concept. Specific weights and thresholds determined during pilot.

Fraud Prevention

How We Detect Manipulation

3 Fraud Patterns We Watch:

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Circular Wallet Flows

Same money moving between accounts to inflate transaction volume

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Synthetic Orders

Fake marketplace orders with no actual delivery

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Velocity Anomalies

Unusual spike in transactions at odd hours

How We Respond:

- Flag for human review
- Require additional verification
- Score penalty applied

Compliance Roadmap

Path to Production

Phase 1

Sandbox (Month 1-2)

- Anonymized data only
- Internal testing
- Manual review for all decisions
- SBV regulatory consultation

→

Phase 2

Pilot (Month 3-4)

- Live data with full consent
- Automated decisions (80%)
- Human review for exceptions
- Compliance audit trail

→

Phase 3

Production (Month 5-6)

- Full automated processing
- Real-time fraud detection
- Complete audit compliance
- Scale to 100+ merchants

The Hacker Advantage

Why This Works

1. Data Fusion Engine

Combined 3 data streams not designed to talk. Normalized across formats, timing, currencies.

2. Vietnam-Specific Heuristics

Tết spikes, school cycles, monsoon disruption, market day rhythms.

3. Explainable Rules

Not a black box. Merchants see WHY their score changed.

Qwen's Role

Where AI adds value

- Parse and categorize transaction descriptions in Vietnamese
- Detect anomalies in spending patterns
- Generate explainable score summaries
- Predict cash flow for loan sizing

Where human cleverness matters more

- Design credit signals that predict default
- Build seasonal adjustments for Vietnam
- Create thresholds for informal businesses
- Balance approval rate vs. default rate

The Business Model

For Shinhan

Revenue

- Loan origination fees
- Interest income: Market-competitive rate
- Deposit growth: SME accounts
- Cross-sell: Insurance, payment processing

Unit Economics

- Unit economics TBD based on early product launch
- Focus: prove the model works first
- Target: prove model performance at acceptable loss rate

The Loan Product

What We Offer

Loan Range	Interest Rate	Term Length	Approval Time
$200 - $5,000	**15-18% APR**	**3, 6, or 12 months**	**72 hours**

Origination Fee	Late Payment	Prepayment	Min. History
1%	**1.5%/mo**	**No penalty**	**90 days**

Illustrative terms for demonstration. Final terms determined during pilot.

The Flywheel

More merchants → More data captured

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Better scoring → Higher approval rates

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Better underwriting → Lower defaults

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Better portfolio → Lower cost of capital

↓

Lower rates → More demand

This data asset compounds over time.

Competitive Position

	Traditional Bank	Existing Alternative Lenders	Shinhan Credit Connect
Data used	Tax statements	One channel (e.g., Shopee, or MISA)	**POS + Marketplace + Wallet COMBINED**
Approval time	5-7 days	5 minutes to 48 hours	**Target: fastest practical**
Min. history	6 months	3 months	**90 days minimum history**
Data breadth	Narrow	Medium	**We combine POS + marketplace + e-wallet data**
SME coverage	Limited	Growing	**Focus: underserved segment**

Sources: MISA Lending (misa.vn), Shopee SEasy (TPBank partnership), Validus Vietnam, Techcombank ShopCash

Early Product Launch Plan

Phase 1 (M1-2):
20 POS merchants
→ data quality, score distribution

Phase 2 (M3-4):
50 digital sellers
→ marketplace data, hybrid scoring

Phase 3 (M5-6):
Full credit product
→ disbursement, defaults, repeat

Success Metric: prove data quality + model signal

What This Is

We haven't launched a production credit product yet.
We don't have live API integrations yet.
We don't have a backtested model yet.

What we HAVE built:

- Working early system showing data → score concept
- Data fusion pipeline architecture
- Clear path to 6-month early product launch

This is an early product proposal with a defined roadmap.

The Ask

What we need

- Access to SOL merchant infrastructure
- Partnership with 1-2 POS providers
- API access to anonymized transaction data
- **6 months** to prove the model

What you get

- Working **alternative credit scoring prototype**
- Path to serving underserved SMEs
- Data asset that compounds over time
- Team that understands both **data engineering** and **credit risk**

The Close

Every transaction is a **data point**.

Every data point is a **credit signal**.

Every credit signal is a **business that can grow**.

We built the credit bureau Vietnam's SMEs deserve.

Appendix: Research Sources

Market Context

Source	What It Verifies
Vietnam News - Digital Loans Open New Doors for SMEs	SMEs struggle to access bank credit; digital lending growing
Techcombank - MISA Lending	Existing bank alternative-data lending product
MISA Lending - MB Partnership	Digital SME lending via accounting software

Existing Competitor Products

Source	What It Verifies
Shopee Seller Education - SEasy	Shopee seller financing product exists
Validus Vietnam	Active SME lender in Vietnam
Techcombank MerchantOne	Bank merchant acceptance product
Techcombank SoftPOS	Bank SoftPOS product
Techcombank ShopCash	Bank merchant lending product